PROJECTE HOSPITAL DE BLANES

DOCUMENT FINAL D'INSTAL·LACIÓ

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ÍNDEX

REQUISITS TÈCNICS

Objectiu principal: L'objectiu del disseny és garantir el funcionament ininterromput (24x7) dels serveis assistencials de l'hospital (accés a historials clínics i gestió de visites) i assegurar la protecció de les dades de caràcter confidencial.

Arquitectura de dos nodes (Actiu-Passiu): Es proposa un sistema basat en programari lliure (PostgreSQL i Pgpool-II) estructurat en dues ubicacions físiques diferents:

- **Node 1 (Local/Actiu):** Maquina Principal

- **Node 2 (Cloud/Passiu):** Replica

- **Gestió d'errors (Failover automàtic):** S'utilitza un component intermig (Pgpool-II) que supervisa contínuament l'estat d'ambdós servidors. Si el servidor de l'hospital deixa de funcionar, el sistema redirigeix automàticament tot el trànsit cap al servidor del núvol.

Estratègia de Còpies de Seguretat: Com a complement a l'alta disponibilitat, s'estableix un procés automàtic de còpies de seguretat diàries. Aquestes es guarden de manera local (mantenint les últimes versions) i també s'envien xifrades a un repositori extern.

Aplicacions:

- SO Linux Ubuntu 24.04.3 LTS
- PostgreSQL v14

Llibreries:

- psycopg2
- faker
- random
- datetime
- pyinstaller
- cryptography

INSTAL·LACIONS PRÈVIES

1. Instal·lem el sistema gestor de base de dades al servidor

```
vicifu007@alexplanas:~$ sudo apt install postgresql
[sudo] contraseña para vicifu007:
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
postgresql ya está en su versión más reciente (16+257build1.1).
Los paquetes indicados a continuación se instalaron de forma automática y ya
 no son necesarios.
  libllvm19 libwoff1 linux-headers-6.14.0-33-generic
  linux-hwe-6.14-headers-6.14.0-33 linux-hwe-6.14-tools-6.14.0-33
  linux-image-6.14.0-33-generic linux-modules-6.14.0-33-generic
  linux-modules-extra-6.14.0-33-generic linux-tools-6.14.0-33-generic
Utilice «sudo apt autoremove» para eliminarlos.
0 actualizados, 0 nuevos se instalarán, 0 para eliminar y 188 no actualizado
s.
```

2. Utilitzem l'usuari postgres i li creem una contrasenya

```
vicifu007@alexplanas:~$ su postgres
Contraseña:
postgres@alexplanas:/home/vicifu007$
```

CREACIÓ DE LA BASE DE DADES

1. Creem la base de dades del nostre hospital

```
postgres@alexplanas:/home/vicifu007$ CREATE DATABASE hospital_santa_paciencia
```

2. Importem el script de l'estructura (taules, atributs) i l'executem: enllaç

```
postgres@alexplanas:/home/vicifu007$ CREATE SCHEMA IF NOT EXISTS hospital;

SET search_path TO hospital;

CREATE TABLE IF NOT EXISTS TREBALLADOR (
id_empleat SERIAL PRIMARY KEY,
dni VARCHAR(9) NOT NULL UNIQUE,
nom VARCHAR(50),
cognom VARCHAR(100),
telefon VARCHAR(15) NOT NULL,
direccio VARCHAR(150)
);

CREATE TABLE IF NOT EXISTS MEDIC (
id_empleat INT PRIMARY KEY,
especialitat VARCHAR(50),
estudi VARCHAR(100),
cv TEXT,
FOREIGN KEY (id_empleat) REFERENCES TREBALLADOR(id_empleat)
);
```

CREACIÓ DE ROLS

1. Importem ara el script dels rols, usuaris, grant, revoke i l'executem: enllaç

```
postgres@alexplanas:/home/vicifu007$ -- =====================================
=======================
-- CREACIÓ DE ROLS
-- ============================================================

CREATE ROLE dba;
CREATE ROLE administracio;
CREATE ROLE medic;
CREATE ROLE infermeria;
CREATE ROLE vari;
CREATE ROLE zelador;

-- ============================================================
-- DBA
-- ============================================================

GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA hospital TO dba;
GRANT ALL PRIVILEGES ON ALL SEQUENCES IN SCHEMA hospital TO dba;
GRANT ALL PRIVILEGES ON ALL FUNCTIONS IN SCHEMA hospital TO dba;

-- ============================================================
-- ADMINISTRACIO
-- ============================================================

GRANT SELECT, INSERT, UPDATE ON hospital.PACIENT   TO administracio;
GRANT SELECT, INSERT, UPDATE ON hospital.RESERVA   TO administracio;
GRANT SELECT, INSERT, UPDATE ON hospital.HABITACIO TO administracio;
GRANT SELECT, INSERT, UPDATE ON hospital.PLANTA    TO administracio;
GRANT SELECT ON hospital.TREBALLADOR TO administracio;

-- ============================================================
-- MEDIC
-- ============================================================

GRANT SELECT, INSERT, UPDATE ON hospital.VISITA   TO medic;
GRANT SELECT, INSERT, UPDATE ON hospital.RECEPTE  TO medic;
GRANT SELECT, INSERT, UPDATE ON hospital.OPERACIO TO medic;

GRANT SELECT ON hospital.PACIENT    TO medic;
GRANT SELECT ON hospital.MEDICAMENT TO medic;
```

SSL

1. Instal·lem el openssl

```
vicifu007@alexplanas:~$ sudo apt update
sudo apt install openssl -y
[sudo] contraseña para vicifu007:
Obj:1 http://es.archive.ubuntu.com/ubuntu noble InRelease
Des:2 http://es.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Des:3 http://es.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Des:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Des:5 http://es.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [
2.007 kB]
Des:6 http://es.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [
355 kB]
Des:7 http://es.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components
 [177 kB]
Des:8 http://es.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Pack
ages [3.202 kB]
Des:9 http://es.archive.ubuntu.com/ubuntu noble-updates/restricted Translatio
n-en [740 kB]
Des:10 http://es.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packa
ges [1.690 kB]
Des:11 http://es.archive.ubuntu.com/ubuntu noble-updates/universe Translation
-en [328 kB]
Des:12 http://es.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Compo
nents [386 kB]
Des:13 http://es.archive.ubuntu.com/ubuntu noble-updates/multiverse Translati
on-en [10,9 kB]
Des:14 http://es.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Com
ponents [940 B]
Des:15 http://es.archive.ubuntu.com/ubuntu noble-backports/main amd64 Compone
nts [5.744 B]
Des:16 http://es.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Com
```

2. Creem el certificat i la clau privada

```
vicifu007@alexplanas:~$ sudo openssl req -x509 -nodes -days 365 -newkey rsa:4
096 \
  -keyout /etc/postgresql/ssl/server.key \
  -out /etc/postgresql/ssl/server.crt \
  -subj "/CN=192.168.56.103"
..+....+..+.+........+.+.....+......+.......+..++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++*.......+.+......+..+......+.+.....+......
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..+.....+..........+.........+.....+..........+.................+...+.+......
+...+..+.............+........+...+.......+........+.+.....+.+............+..
```

3. Donem propietat a l'usuari de la base de dades i que només ell pugui llegir la clau privada

4. I afegim que el certificat pugui ser llegit per tothom

```
vicifu007@alexplanas:~$ sudo chown postgres:postgres /etc/postgresql/ssl/serv
er.key /etc/postgresql/ssl/server.crt
vicifu007@alexplanas:~$ sudo chmod 600 /etc/postgresql/ssl/server.key
vicifu007@alexplanas:~$ sudo chmod 644 /etc/postgresql/ssl/server.crt
```

5. Creem el fitxer per activar el SSL en postgres, modifiquem les següents línies:

```
vicifu007@alexplanas:~$ sudo nano /etc/postgresql/16/main/postgresql.conf
vicifu007@alexplanas:~$ sudo systemctl restart postgresql
```

```
  GNU nano 7.2        /etc/postgresql/16/main/postgresql.conf
# -----------------------------
# PostgreSQL configuration file
# -----------------------------
#
# This file consists of lines of the form:
#
#   name = value
#
# (The "=" is optional.)  Whitespace may be used.  Comments are introduced w>
# "#" anywhere on a line.  The complete list of parameter names and allowed
# values can be found in the PostgreSQL documentation.
#
# The commented-out settings shown in this file represent the default values.
# Re-commenting a setting is NOT sufficient to revert it to the default valu>
# you need to reload the server.
#
# This file is read on server startup and when the server receives a SIGHUP
# signal.  If you edit the file on a running system, you have to SIGHUP the
# server for the changes to take effect, run "pg_ctl reload", or execute
# "SELECT pg_reload_conf()".  Some parameters, which are marked below,
# require a server shutdown and restart to take effect.
#
# Any parameter can also be given as a command-line option to the server, e.>
# "postgres -c log_connections=on".  Some parameters can be changed at run t>
# with the "SET" SQL command.
#
# Memory units:  B  = bytes            Time units:  us  = microseconds
#                kB = kilobytes                     ms  = milliseconds
#                MB = megabytes                     s   = seconds
#                GB = gigabytes                     min = minutes
#                TB = terabytes                     h   = hours
#                                                   d   = days
```

```
# - SSL -

ssl = on
#ssl_ca_file = ''
ssl_cert_file = '/etc/postgresql/ssl/server.crt'
#ssl_crl_file = ''
#ssl_crl_dir = ''
ssl_key_file = '/etc/postgresql/ssl/server.key'
#ssl_ciphers = 'HIGH:MEDIUM:+3DES:!aNULL' # allowed SSL ciphers
#ssl_prefer_server_ciphers = on
#ssl_ecdh_curve = 'prime256v1'
#ssl_min_protocol_version = 'TLSv1.2'
#ssl_max_protocol_version = ''
#ssl_dh_params_file = ''
#ssl_passphrase_command = ''
#ssl_passphrase_command_supports_reload = off
```

6. Reinciem postgres per guardar la configuració

7. Creem el directori pels certificats i assignem propietat a l'usuari postgres

```
vicifu007@alexplanas:~$ sudo mkdir -p /etc/postgresql/ssl
vicifu007@alexplanas:~$ sudo chown postgres:postgres /etc/postgresql/ssl
vicifu007@alexplanas:~$ sudo chmod 700 /etc/postgresql/ssl
vicifu007@alexplanas:~$
```

8. Creem un fitxer per generar nous certificats cada 365 dies

```
vicifu007@alexplanas:~$ sudo chmod +x /usr/local/bin/renovar-postgres-ssl.sh
```

9. Enganxarem el següent script: enllaç

```
  GNU nano 7.2          /usr/local/bin/renovar-postgres-ssl.sh
#!/bin/bash

DIES=365
RUTA="/etc/postgresql/ssl"
IP_VM="192.168.56.103"

openssl req -x509 -nodes -days $DIES -newkey rsa:4096 \
  -keyout $RUTA/server.key \
  -out $RUTA/server.crt \
  -subj "/CN=$IP_VM"

chown postgres:postgres $RUTA/server.key $RUTA/server.crt
chmod 600 $RUTA/server.key
chmod 644 $RUTA/server.crt

systemctl restart postgresql
```

10. Accedim al crontab per automatitzar els backups

```
vicifu007@alexplanas:~$ sudo crontab -e
```

11. Afegim la següent línia al final:

```
  GNU nano 7.2                    /tmp/crontab.b9JnTx/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 0 1 1 * /usr/local/bin/renovar-postgres-ssl.sh >> /var/log/renovacio-ssl-p>
00 03 * * * /home/el_teu_usuari/scripts/backup_bd.sh
```

12. Verifiquem que el ssl estigui activat:

```
vicifu007@alexplanas:~$ sudo -u postgres psql
psql (16.13 (Ubuntu 16.13-0ubuntu0.24.04.1))
Type "help" for help.

postgres=# SHOW ssl;
 ssl
-----
 on
(1 row)
```

DATA MASKING

1. Executem el següent script amb l'usuari postgres per crear les vistes per a vari i zelador: [enllaç]()

```sql
--VISTA VARI

CREATE OR REPLACE VIEW hospital.v_pacient_vari AS
SELECT
id_pacient,
'XXXXX' || RIGHT(dni, 4) AS dni,
'******' || RIGHT(telefon, 3) AS telefon,
LEFT(email, 1) || '*******@' || SPLIT_PART(email, '@', 2) AS email,
nom,
cognom
FROM hospital.pacient;

GRANT SELECT ON hospital.v_pacient_vari TO vari;

--VISTA ZELADOR

CREATE OR REPLACE VIEW hospital.v_pacient_zelador AS
SELECT
id_pacient,
'XXXXX' || RIGHT(dni, 4) AS dni,
'******' || RIGHT(telefon, 3) AS telefon,
LEFT(email, 1) || '*******@' || SPLIT_PART(email, '@', 2) AS email,
nom,
cognom
FROM hospital.pacient;

GRANT SELECT ON hospital.v_pacient_zelador TO zelador;
```

REPLICACIÓ

1. Creem l'usuari de replicació: [enllaç]

```
vicifu007@alexplanas:~$ sudo -u postgres psql -c "CREATE ROLE replica_user RE
PLICATION LOGIN PASSWORD '1234';"
ERROR:  role "replica_user" already exists
```

2. Editem el fitxer de configuració i canviem les següents línies:

```
sudo nano /etc/postgresql/16/main/postgresql.conf
```

```
#------------------------------------------------------------------------------
# REPLICATION
#------------------------------------------------------------------------------

# - Sending Servers -

# Set these on the primary and on any standby that will send replication dat

max_wal_senders = 10            # max number of walsender processes
                                # (change requires restart)
#max_replication_slots = 10     # max number of replication slots
                                # (change requires restart)
#wal_keep_size = 0              # in megabytes; 0 disables
#max_slot_wal_keep_size = -1    # in megabytes; -1 disables
#wal_sender_timeout = 60s       # in milliseconds; 0 disables
#track_commit_timestamp = off   # collect timestamp of transaction commit
                                # (change requires restart)
```

```
#------------------------------------------------------------------------------
# CONNECTIONS AND AUTHENTICATION
#------------------------------------------------------------------------------

# - Connection Settings -

listen_addresses = '*'                  # what IP address(es) to listen on;
                                        # comma-separated list of addresses;
                                        # defaults to 'localhost'; use '*' f
                                        # (change requires restart)
```

```
#------------------------------------------------------------------------------>
# WRITE-AHEAD LOG
#------------------------------------------------------------------------------>

# - Settings -

wal_level = replica                     # minimal, replica, or logical
```

3. Editem el el fitxer d'autenticació:

```
vicifu007@alexplanas:~$ sudo nano /etc/postgresql/16/main/pg_hba.conf
vicifu007@alexplanas:~$ sudo systemctl restart postgresql
```

4. Afegim la següent línia al final:

```
  GNU nano 7.2              /etc/postgresql/16/main/pg_hba.conf
#
# ----------------------------------
# Put your actual configuration here
# ----------------------------------
#
# If you want to allow non-local connections, you need to add more
# "host" records.  In that case you will also need to make PostgreSQL
# listen on a non-local interface via the listen_addresses
# configuration parameter, or via the -i or -h command line switches.




# DO NOT DISABLE!
# If you change this first entry you will need to make sure that the
# database superuser can access the database using some other method.
# Noninteractive access to all databases is required during automatic
# maintenance (custom daily cronjobs, replication, and similar tasks).
#
# Database administrative login by Unix domain socket
local   all             postgres                                peer

# TYPE  DATABASE        USER            ADDRESS                 METHOD

# "local" is for Unix domain socket connections only
local   all             all                                     peer
# IPv4 local connections:
host    all             all             0.0.0.0/0            scram-sha-256
# IPv6 local connections:
host    all             all             ::1/128                 scram-sha-256
# Allow replication connections from localhost, by a user with the
# replication privilege.
local   replication     all                                     peer
host    replication     all             127.0.0.1/32            scram-sha-256
host    replication     all             ::1/128                 scram-sha-256
host    replication      replica_user   192.168.56.0/24         md5
```

5. Reiniciem per aplicar canvis

6. Instal·lem postgres a la nova màquina:

```
bd-replica@bd-replica-VirtualBox:~$ sudo apt update
[sudo] contraseña para bd-replica:
Des:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Obj:2 http://archive.ubuntu.com/ubuntu noble InRelease
Des:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Des:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Des:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1.69
3 kB]
Des:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2.007
kB]
Des:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [267
kB]
Des:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [42
```

```
bd-replica@bd-replica-VirtualBox:~$ sudo apt install postgresql-16 postgresql-cl
ient-16 -y
```

7. Aturem el servei i buidem el directori:

```
bd-replica@bd-replica-VirtualBox:~$ sudo systemctl stop postgresql
[sudo] contraseña para bd-replica:
bd-replica@bd-replica-VirtualBox:~$ sudo rm-rf /var/lib/postgresql/16/main/*
```

8. Sincronitzem la base de dades des de la màquina principal

```
vicifu007@alexplanas:~$ sudo -u postgres pg_basebackup -h 192.168.56.103 -D
/var/lib/postgresql/16/main/ -U replica_user -P -R
Password:
```

9. Tornem a posar en marxa a la replica:

```
bd-replica@bd-replica-VirtualBox:~$ sudo systemctl start postgresql
```

10. Verifiquem que s'ha sincronitzat correctament creant el següent al node 1

```
vicifu007@alexplanas:~$       sudo -u postgres psql -c "CREATE TABLE comproba
cio_final (id SERIAL PRIMARY KEY, missatge TEXT);"
      sudo -u postgres psql -c "INSERT INTO comprobacio_final (missatge) VALU
ES ('Replicacio funcionant correctament');"
ERROR:  relation "comprobacio_final" already exists
INSERT 0 1
```

11. Mirem des de el node 2 que s'hagi replicat:

```
bd-replica@bd-replica-VirtualBox: ~
bd-replica@bd-replica-VirtualBox:~$ sudo -u postgres psql -c "SELECT * FROM comp
robacio_final;"
[sudo] contraseña para bd-replica:
 id |              missatge
----+------------------------------------
  1 | Replicacio funcionant correctament
(1 row)

bd-replica@bd-replica-VirtualBox:~$ S
```

BACKUPS

1. Creem el fitxer per a la creació de backups:

```
vicifu007@alexplanas:~$
      nano ~/scripts/backup_bd.sh
```

2. Enganxem el següent contingut: enllaç

```
 GNU nano 7.2            /home/vicifu007/scripts/backup_bd.sh
#!/bin/bash
# Script de Backup - Hospital Santa Paciencia

# Configuració
DATE=$(date +%Y%m%d_%H%M%S)
BACKUP_DIR="/var/backups/postgres"
DB_NAME="hospital_santa_paciencia"  # Assegura't que aquest és el nom de la >
KEEP_DAYS=5

# Crear el directori de destí si no existeix
sudo mkdir -p $BACKUP_DIR
sudo chown postgres:postgres $BACKUP_DIR

echo "Iniciant backup de $DB_NAME..."

# Executar el backup
# Nota: fem servir sudo per executar-ho com l'usuari postgres
sudo -u postgres pg_dump $DB_NAME > $BACKUP_DIR/backup_$DATE.sql

# Rotació: Esborrar fitxers de més de 5 dies (Requeriment 3)
sudo find $BACKUP_DIR -type f -mtime +$KEEP_DAYS -name "*.sql" -delete

echo "Backup finalitzat a $BACKUP_DIR/backup_$DATE.sql"
```

3. Comprovem que s'executen els backups i es guarden al directori:

```
vicifu007@alexplanas:~$ chmod +x ~/scripts/backup_bd.sh
vicifu007@alexplanas:~$ sudo ~/scripts/backup_bd.sh
Iniciant backup de hospital_santa_paciencia...
Backup finalitzat a /var/backups/postgres/backup_20260525_173359.sql
vicifu007@alexplanas:~$ ls -lh /var/backups/postgres
total 17M
-rw-r--r-- 1 root root 8,2M may 25 17:33 backup_20260525_173339.sql
-rw-r--r-- 1 root root 8,2M may 25 17:34 backup_20260525_173359.sql
vicifu007@alexplanas:~$ |
```

4. Editem el crontab -e i afegim la següent línia per automatitzar-ho a las 3 a.m.:

```
  GNU nano 7.2                    /tmp/crontab.EoyxXm/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
00 03 * * * /home/vicifu007/scripts/backup_bd.sh
```

RECUPERAR BACKUP

1. Primer s'executa l'script de backup automatitzat per tenir una imatge actual de la base de dades:

```
vicifu007@alexplanas:~$ sudo ~/scripts/backup_bd.sh
Iniciant backup de hospital_santa_paciencia...
Backup finalitzat a /var/backups/postgres/backup_20260526_120523.sql
vicifu007@alexplanas:~$ ls -lh /var/backups/postgres
total 25M
-rw-r--r-- 1 root root 8,2M may 25 17:33 backup_20260525_173339.sql
-rw-r--r-- 1 root root 8,2M may 25 17:34 backup_20260525_173359.sql
-rw-r--r-- 1 root root 8,2M may 26 12:05 backup_20260526_120523.sql
```

2. S'accedeix a la base de dades i s'elimina una taula crítica per simular un error humà o una fallada del sistema:

```
vicifu007@alexplanas:~$ sudo -u postgres psql -d hospital_santa_paciencia
psql (16.14 (Ubuntu 16.14-0ubuntu0.24.04.1))
Type "help" for help.

hospital_santa_paciencia=# DROP TABLE hospital.reserva;
DROP TABLE
hospital_santa_paciencia=# \dt hospital.*
              List of relations
  Schema  |     Name      | Type  |  Owner
----------+---------------+-------+----------
 hospital | aparell_medic | table | postgres
 hospital | assigna       | table | postgres
 hospital | assisteix     | table | postgres
 hospital | habitacio     | table | postgres
 hospital | infermeria    | table | postgres
 hospital | medic         | table | postgres
 hospital | medicament    | table | postgres
 hospital | operacio      | table | postgres
 hospital | pacient       | table | postgres
 hospital | planta        | table | postgres
 hospital | quirofan      | table | postgres
 hospital | recepte       | table | postgres
 hospital | treballador   | table | postgres
 hospital | vari          | table | postgres
 hospital | visita        | table | postgres
(15 rows)
```

3. S'utilitza l'script de restauració total creat prèviament, indicant la ruta del fitxer de backup generat en el pas 1:

```
hospital_santa_paciencia=# \q
vicifu007@alexplanas:~$ sudo ~/scripts/restore_total.sh /var/backups/postgres/backup_20260526_120523.sql
--- Iniciant procés de restauració total ---
Desconnectant usuaris actius...
 pg_terminate_backend
----------------------
(0 rows)

Restaurant dades des de /var/backups/postgres/backup_20260526_120523.sql...
SET
SET
SET
SET
SET
 set_config
------------

(1 row)

SET
SET
SET
SET
psql:/var/backups/postgres/backup_20260526_120523.sql:25: ERROR:  schema "hospital" already exists
ALTER SCHEMA
ALTER SCHEMA
```

4. Es comprova que la taula reserva s'ha tornat a crear amb tot el seu contingut original:

```
 id_reserva | dia_ingres | hora_ingres | dia_sortida | hora_sortida |         motiu          | id_habitacio | id_pacient
| id_quirofan | id_medic
------------+------------+-------------+-------------+--------------+------------------------+--------------+------------
+-------------+----------
          1 | 2026-05-19 | 08:00:00    | 2026-05-21  | 12:00:00     | Ingrés post-operatori |           44 |      11225
|             |       46
          2 | 2026-05-19 | 08:00:00    | 2026-05-24  | 12:00:00     | Ingrés post-operatori |           36 |      34897
|             |       86
          3 | 2026-05-19 | 08:00:00    | 2026-05-20  | 12:00:00     | Ingrés post-operatori |           39 |      12503
|             |       52
          4 | 2026-05-19 | 08:00:00    | 2026-05-23  | 12:00:00     | Ingrés post-operatori |            8 |      24644
|             |        1
          5 | 2026-05-19 | 08:00:00    | 2026-05-21  | 12:00:00     | Ingrés post-operatori |           48 |      47368
|             |       70
(5 rows)
```